

```
: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("traffic_cleaned.csv")

df['date'] = pd.to_datetime(df['date'])

print("Dataset shape:", df.shape)
```

Dataset shape: (122567, 9)

```
total_events = len(df)

pageviews = (df['event'] == 'pageview').sum()
clicks = (df['event'] == 'click').sum()
previews = (df['event'] == 'preview').sum()

pageview_click_rate = (clicks / pageviews) * 100
pageview_preview_rate = (previews / pageviews) * 100

kpis = {
    'Total Events': total_events,
    'Pageviews': pageviews,
    'Clicks': clicks,
    'Previews': previews,
    'Pageview → Click Rate': pageview_click_rate,
    'Pageview → Preview Rate': pageview_preview_rate,
    'Countries': df['country'].nunique(),
    'Cities': df['city'].nunique(),
    'Artists': df['artist'].nunique(),
    'Tracks': df['track'].nunique(),
    'Links': df['linkid'].nunique()
}

kpis
```

```
{ 'Total Events': 122567,  
  'Pageviews': np.int64(73360),  
  'Clicks': np.int64(32499),  
  'Previews': np.int64(16708),  
  'Pageview → Click Rate': np.float64(44.30070883315158),  
  'Pageview → Preview Rate': np.float64(22.77535441657579),  
  'Countries': 211,  
  'Cities': 11993,  
  'Artists': 2419,  
  'Tracks': 3562,  
  'Links': 3839}
```

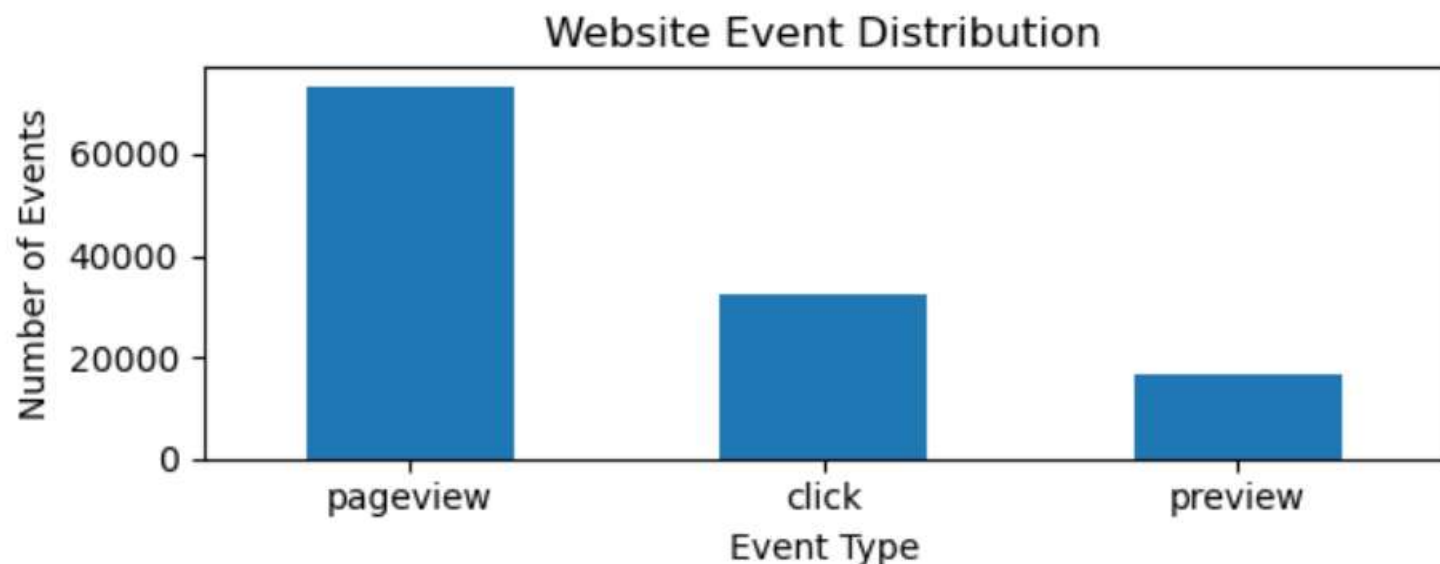
```
event_counts = df['event'].value_counts()

plt.figure(figsize=(6,2.5))

event_counts.plot(kind='bar')

plt.title('Website Event Distribution')
plt.xlabel('Event Type')
plt.ylabel('Number of Events')
plt.xticks(rotation=0)

plt.tight_layout()
plt.show()
```



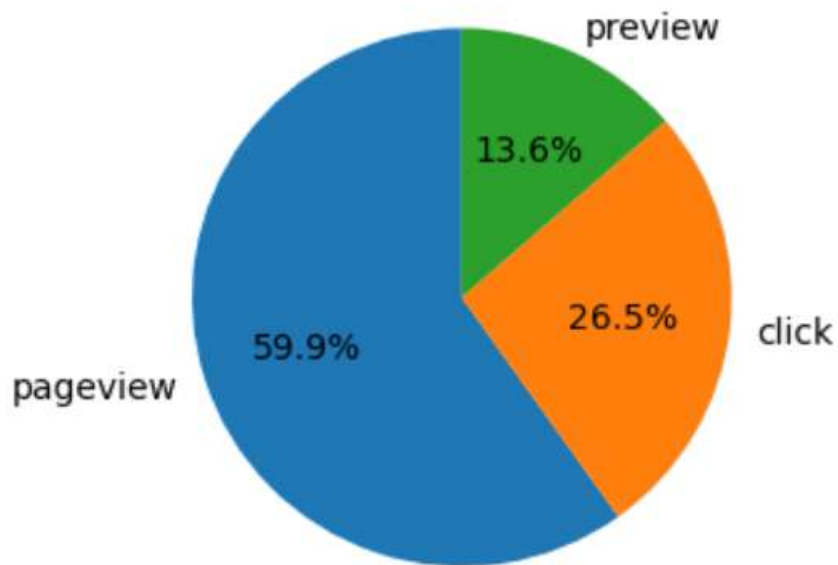
```
] plt.figure(figsize=(3.5,3.5))

plt.pie(
    event_counts.values,
    labels=event_counts.index,
    autopct='%1.1f%%',
    startangle=90
)

plt.title('Website Events - Percentage Distribution')

plt.show()
```

Website Events - Percentage Distribution



```
daily_traffic = df.groupby('date').size()

plt.figure(figsize=(12, 5))

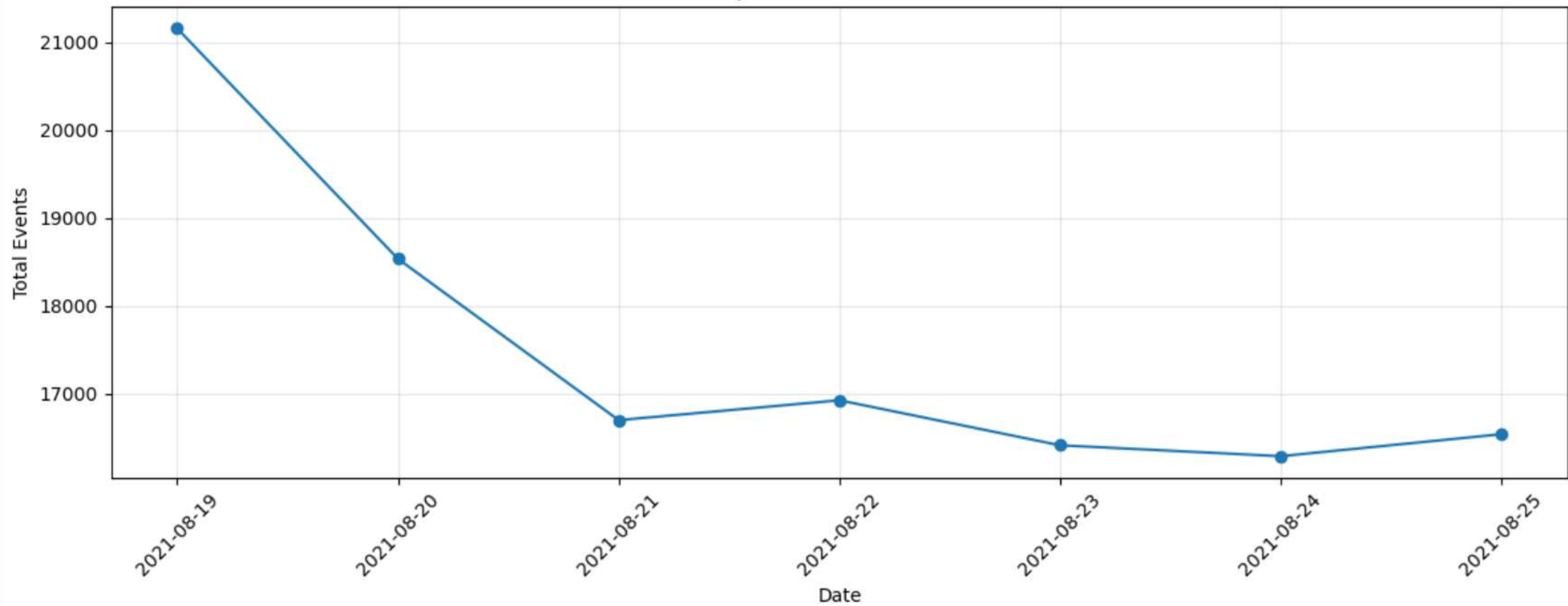
plt.plot(
    daily_traffic.index,
    daily_traffic.values,
    marker='o'
)

plt.title('Daily Website Traffic Trend')
plt.xlabel('Date')
plt.ylabel('Total Events')

plt.xticks(rotation=45)
plt.grid(alpha=0.3)

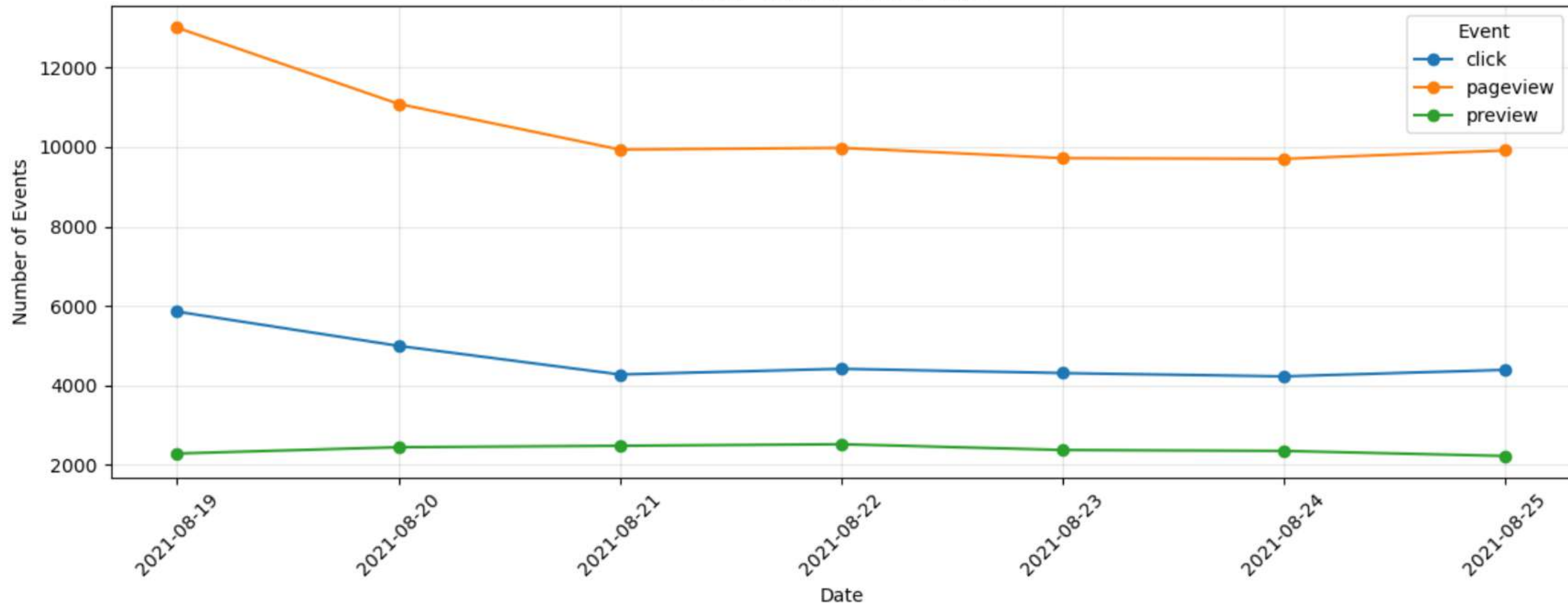
plt.tight_layout()
plt.show()
```

Daily Website Traffic Trend



```
daily_event = (  
    df.groupby(['date', 'event'])  
        .size()  
        .unstack(fill_value=0)  
)  
  
plt.figure(figsize=(12, 6))  
  
for event in daily_event.columns:  
    plt.plot(  
        daily_event.index,  
        daily_event[event],  
        marker='o',  
        label=event  
    )  
  
plt.title('Daily Traffic by Event Type')  
plt.xlabel('Date')  
plt.ylabel('Number of Events')  
  
plt.xticks(rotation=45)  
plt.legend(title='Event')  
plt.grid(alpha=0.3)  
  
plt.tight_layout()  
plt.show()
```

Daily Traffic by Event Type



```
] country_traffic = df['country'].value_counts().head(10)
```

```
plt.figure(figsize=(6, 3))
```

```
country_traffic.sort_values().plot(kind='barh')
```

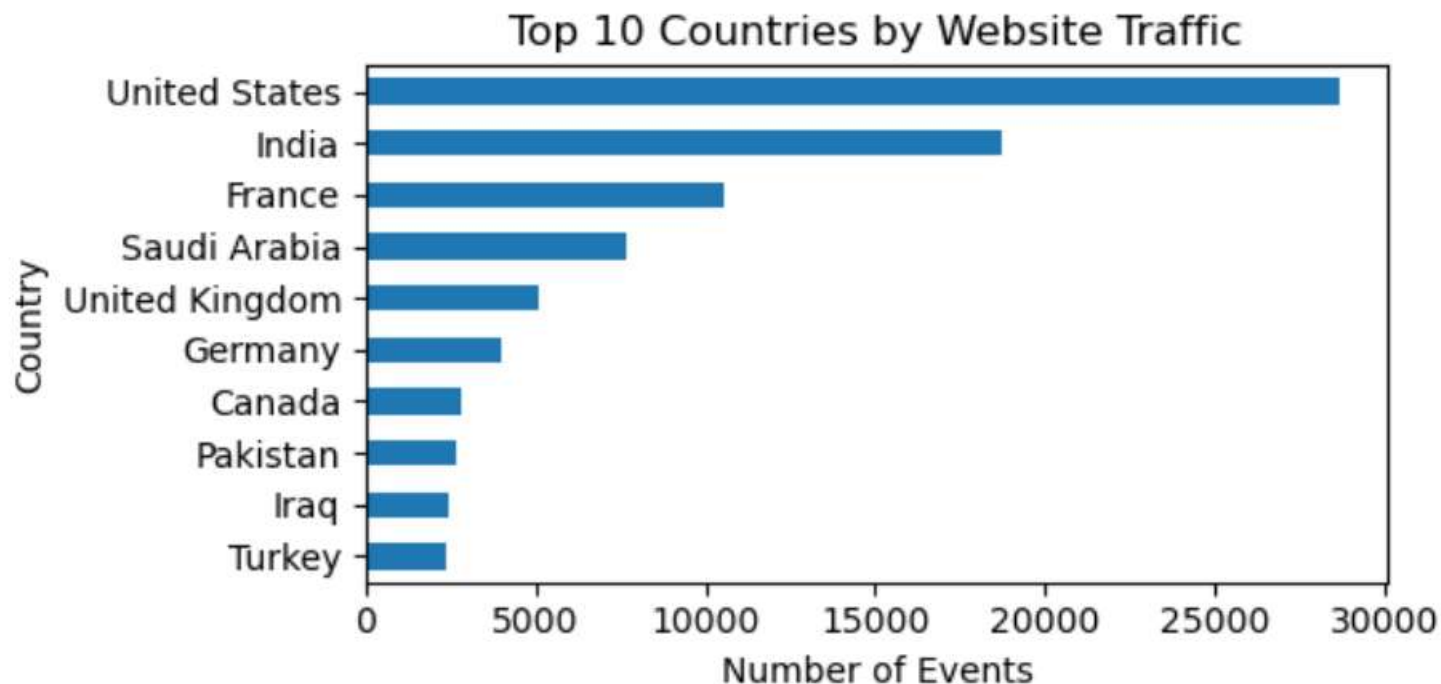
```
plt.title('Top 10 Countries by Website Traffic')
```

```
plt.xlabel('Number of Events')
```

```
plt.ylabel('Country')
```

```
plt.tight_layout()
```

```
plt.show()
```



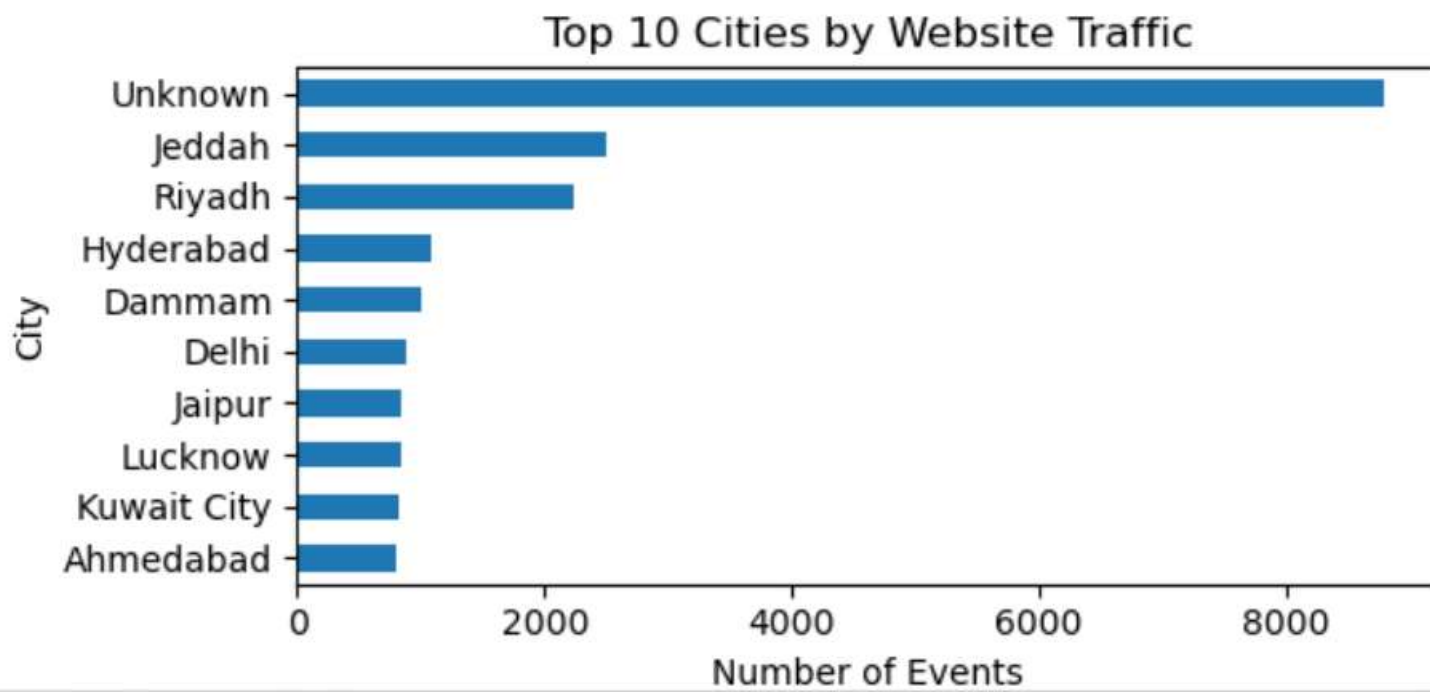
```
9]: city_traffic = df['city'].value_counts().head(10)

plt.figure(figsize=(6,3))

city_traffic.sort_values().plot(kind='barh')

plt.title('Top 10 Cities by Website Traffic')
plt.xlabel('Number of Events')
plt.ylabel('City')

plt.tight_layout()
plt.show()
```



```
] artist_traffic = df['artist'].value_counts().head(10)
```

```
plt.figure(figsize=(8,3))
```

```
artist_traffic.sort_values().plot(kind='barh')
```

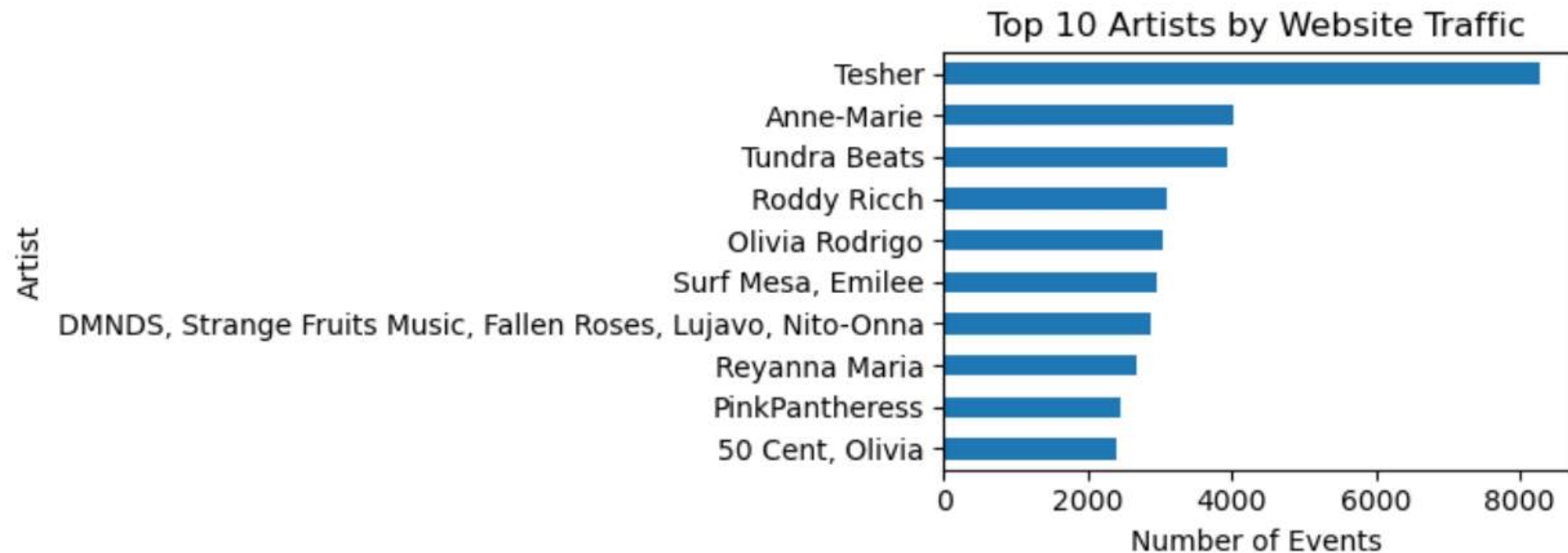
```
plt.title('Top 10 Artists by Website Traffic')
```

```
plt.xlabel('Number of Events')
```

```
plt.ylabel('Artist')
```

```
plt.tight_layout()
```

```
plt.show()
```



```
3]: track_traffic = df['track'].value_counts().head(10)
```

```
plt.figure(figsize=(8,3))
```

```
track_traffic.sort_values().plot(kind='barh')
```

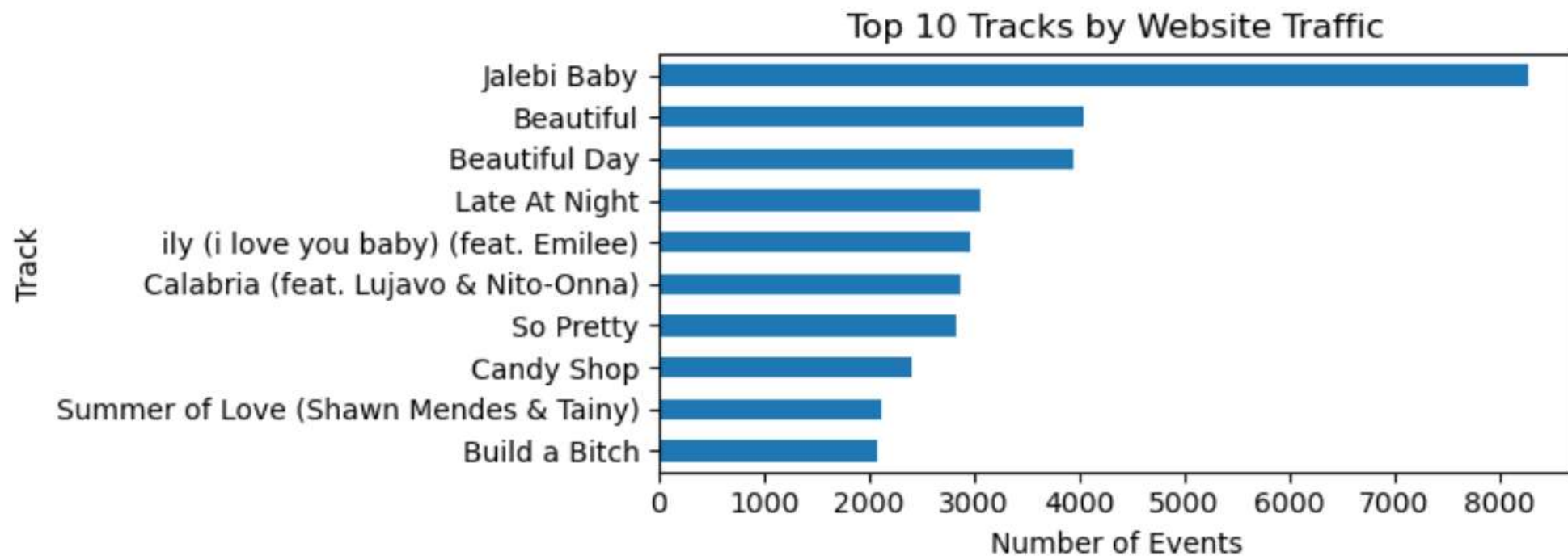
```
plt.title('Top 10 Tracks by Website Traffic')
```

```
plt.xlabel('Number of Events')
```

```
plt.ylabel('Track')
```

```
plt.tight_layout()
```

```
plt.show()
```



```

link_traffic = df['linkid'].value_counts().head(10)

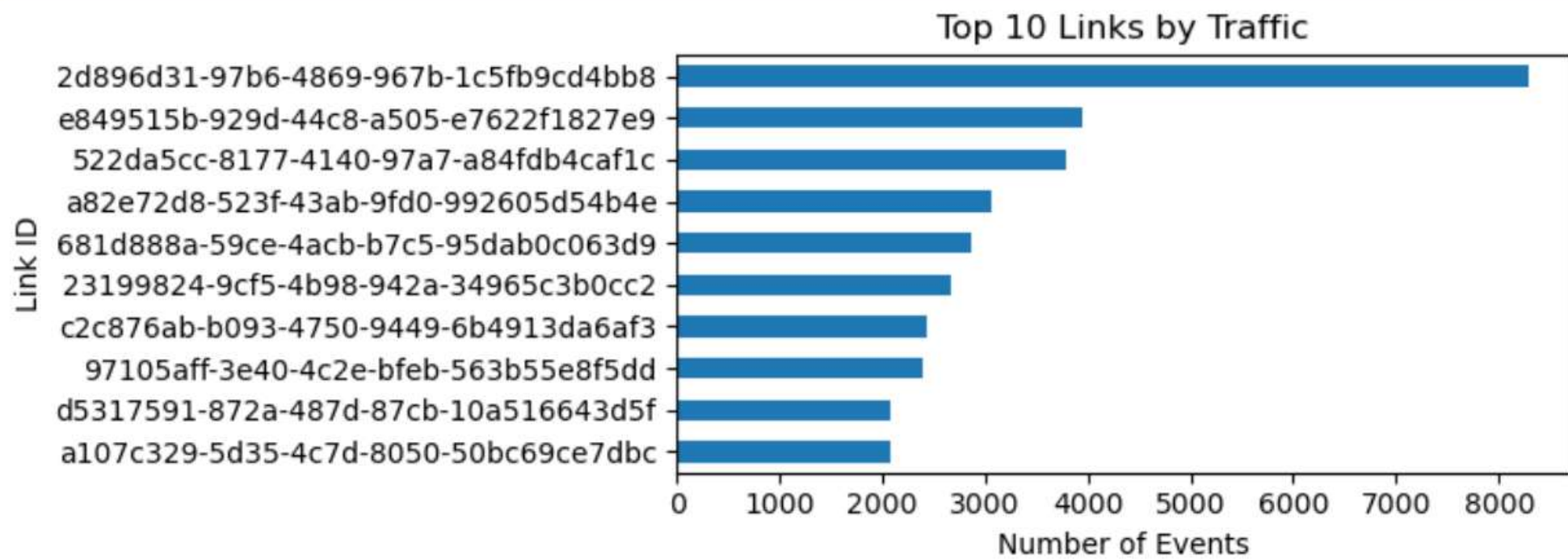
plt.figure(figsize=(8,3))

link_traffic.sort_values().plot(kind='barh')

plt.title('Top 10 Links by Traffic')
plt.xlabel('Number of Events')
plt.ylabel('Link ID')

plt.tight_layout()
plt.show()

```



```
: country_event = pd.crosstab(  
    df['country'],  
    df['event']  
)  
|  
top_countries = df['country'].value_counts().head(10).index  
  
country_event_top = country_event.loc[top_countries]  
  
country_event_top
```

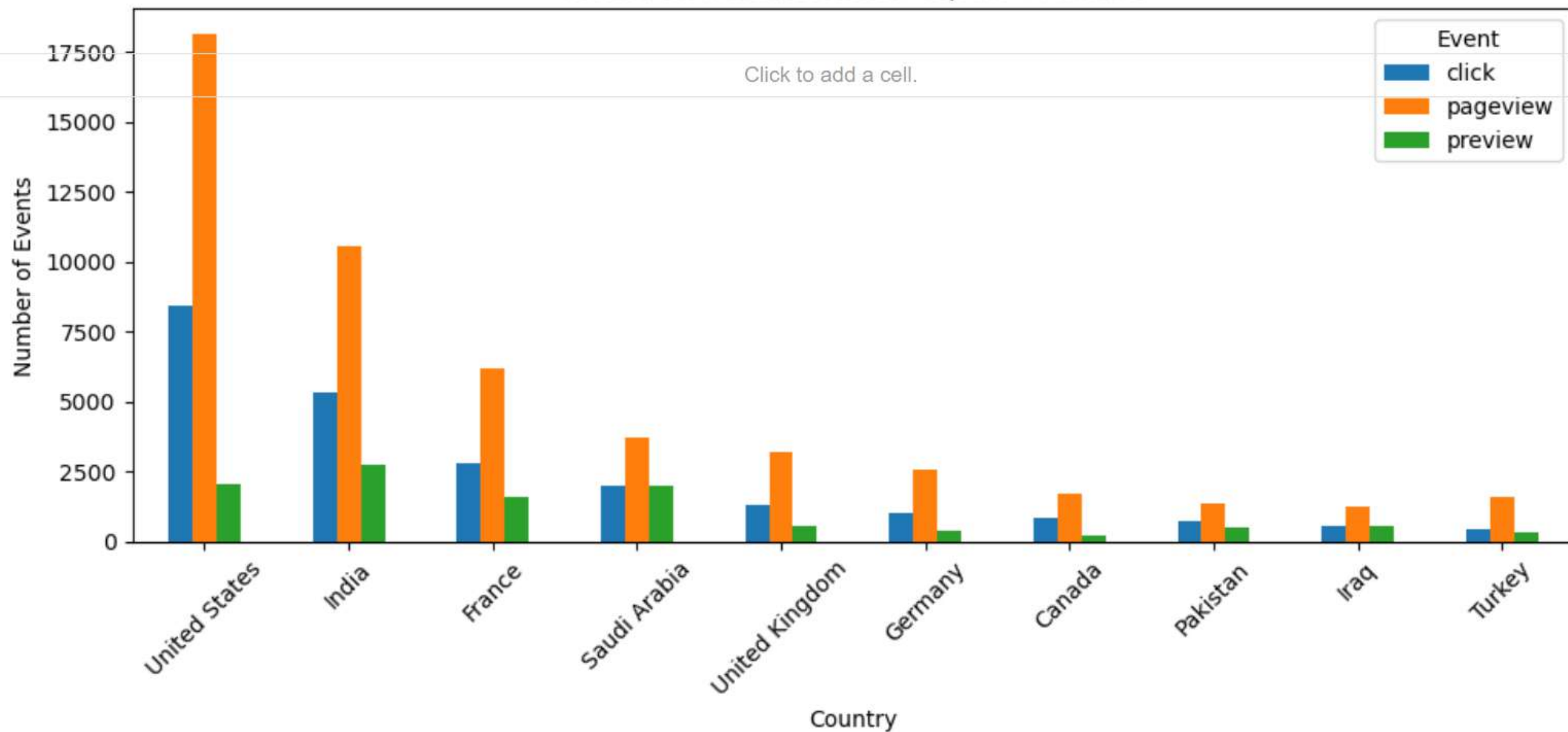
:

event	click	pageview	preview
country			
United States	8446	18131	2087
India	5360	10565	2764
France	2792	6181	1592
Saudi Arabia	1976	3712	1994
United Kingdom	1307	3218	570
Germany	1030	2585	402
Canada	840	1702	242
Pakistan	754	1382	497
Iraq	567	1285	592
Turkey	466	1598	335

```
country_event_top.plot(  
    kind='bar',  
    figsize=(12,6)  
)
```

```
plt.title('Event Performance Across Top 10 Countries')  
plt.xlabel('Country')  
plt.ylabel('Number of Events')  
plt.xticks(rotation=45)  
  
plt.legend(title='Event')  
  
plt.tight_layout()  
plt.show()
```

Event Performance Across Top 10 Countries



```
] funnel_data = pd.Series({
    'Pageviews': pageviews,
    'Previews': previews,
    'Clicks': clicks
})

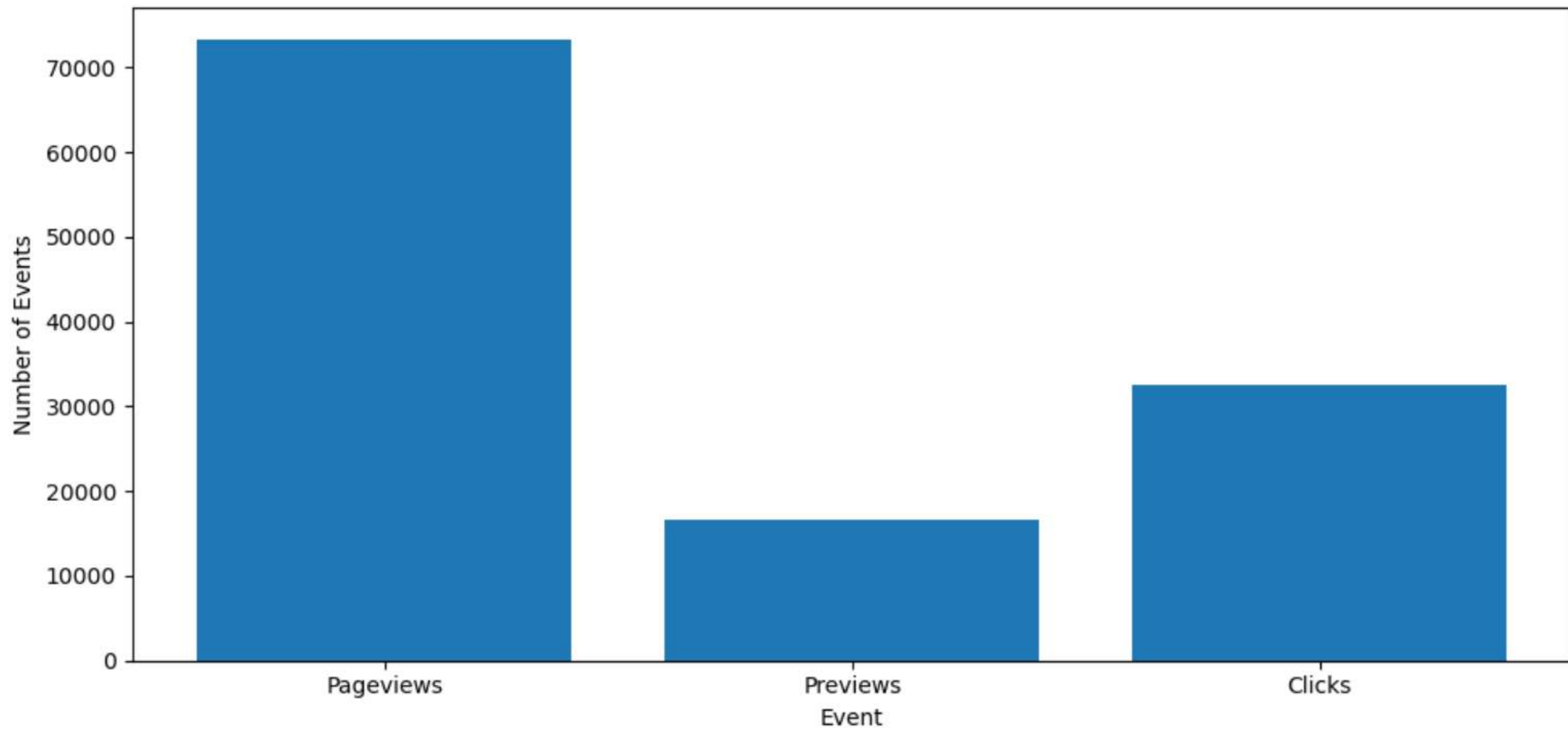
plt.figure(figsize=(10,5))

plt.bar(
    funnel_data.index,
    funnel_data.values
)

plt.title('Website Event Funnel')
plt.xlabel('Event')
plt.ylabel('Number of Events')

plt.tight_layout()
plt.show()
```

Website Event Funnel



```
] : country_metrics = pd.crosstab(  
    df['country'],  
    df['event']  
)  
  
country_metrics['click_rate'] = (  
    country_metrics['click'] /  
    country_metrics['pageview'].replace(0, np.nan)  
) * 100
```

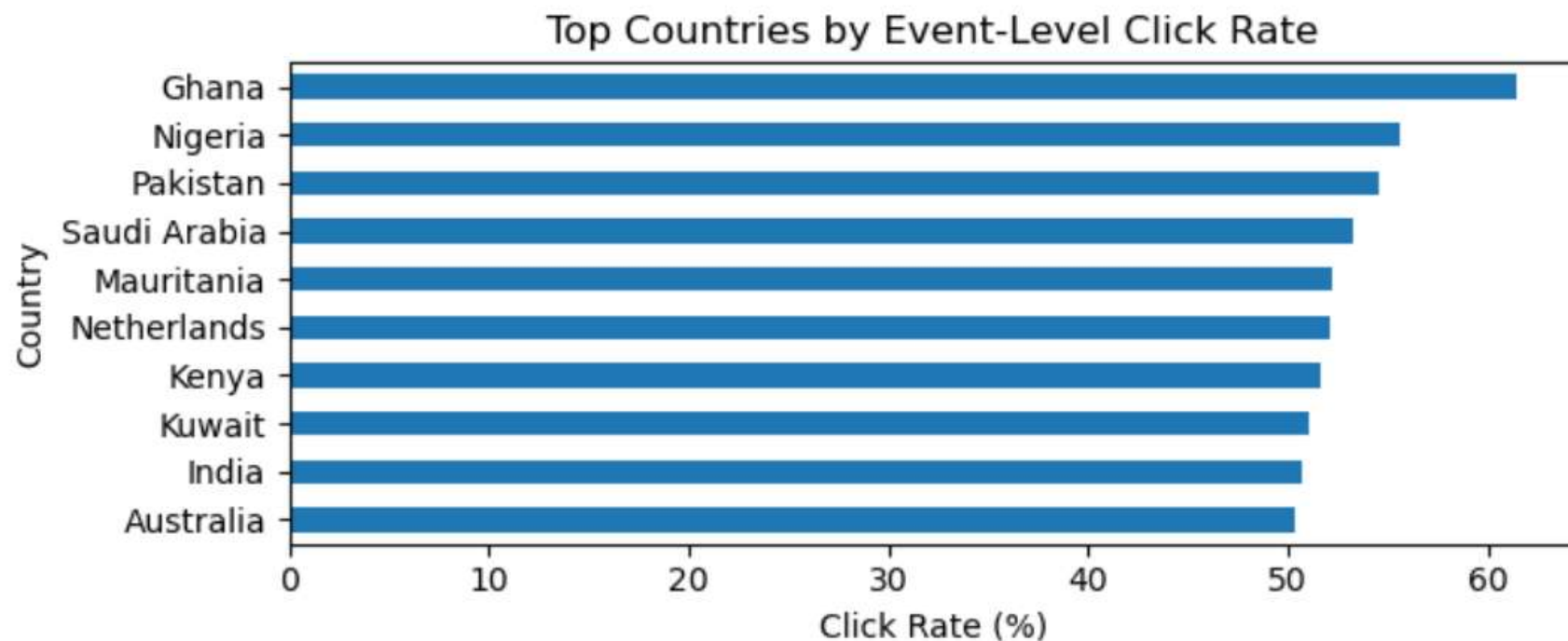
```
] : country_engagement = country_metrics[  
    country_metrics['pageview'] >= 100  
>].sort_values(  
    'click_rate',  
    ascending=False  
)>.head(10)
```

```
plt.figure(figsize=(7, 3))

country_engagement['click_rate'].sort_values().plot(
    kind='barh'
)

plt.title('Top Countries by Event-Level Click Rate')
plt.xlabel('Click Rate (%)')
plt.ylabel('Country')

plt.tight_layout()
plt.show()
```



```
7]: artist_metrics = pd.crosstab(  
    df['artist'],  
    df['event']  
)  
  
artist_metrics['click_rate'] = (  
    artist_metrics['click'] /  
    artist_metrics['pageview'].replace(0, np.nan)  
)
```

```
8]: artist_metrics['click_rate'] *= 100
```

```
9]: artist_engagement = artist_metrics[  
    artist_metrics['pageview'] >= 50  
>].sort_values(  
    'click_rate',  
    ascending=False  
)>.head(10)
```

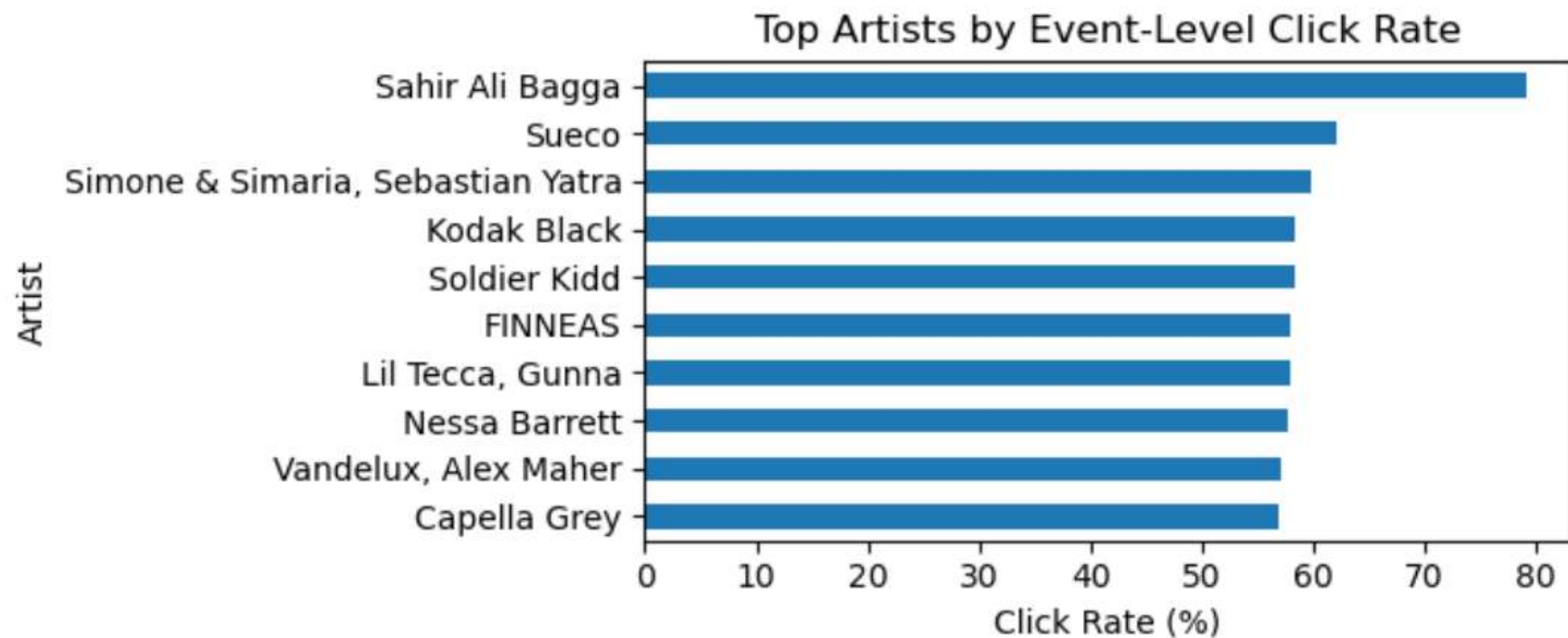
```
]:
```

```
plt.figure(figsize=(7, 3))

artist_engagement['click_rate'].sort_values().plot(
    kind='barh'
)

plt.title('Top Artists by Event-Level Click Rate')
plt.xlabel('Click Rate (%)')
plt.ylabel('Artist')

plt.tight_layout()
plt.show()
```



```
link_metrics = pd.crosstab(
    df['linkid'],
    df['event']
)

link_metrics['click_rate'] = (
    link_metrics['click'] /
    link_metrics['pageview'].replace(0, np.nan)
) * 100

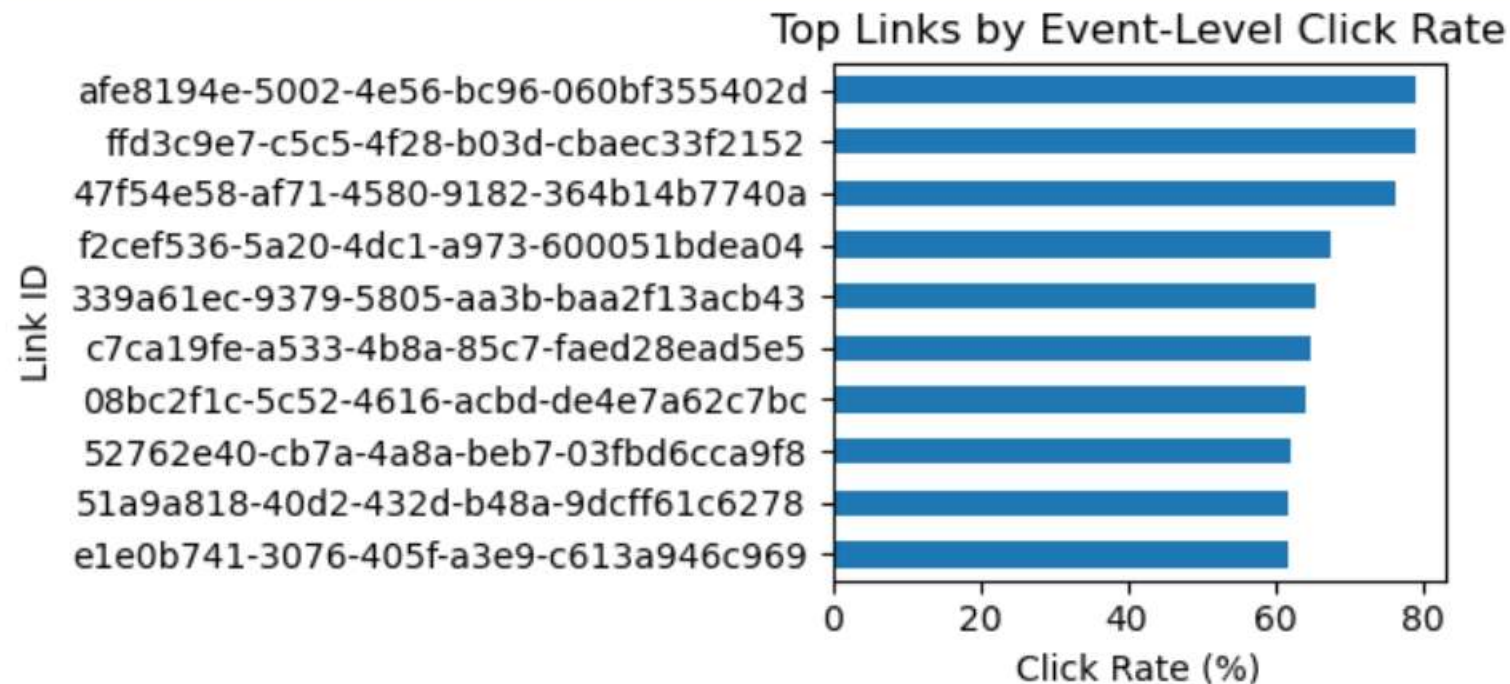
link_engagement = link_metrics[
    link_metrics['pageview'] >= 20
].sort_values(
    'click_rate',
    ascending=False
).head(10)
```

```
plt.figure(figsize=(6,3))

link_engagement['click_rate'].sort_values().plot(
    kind='barh'
)

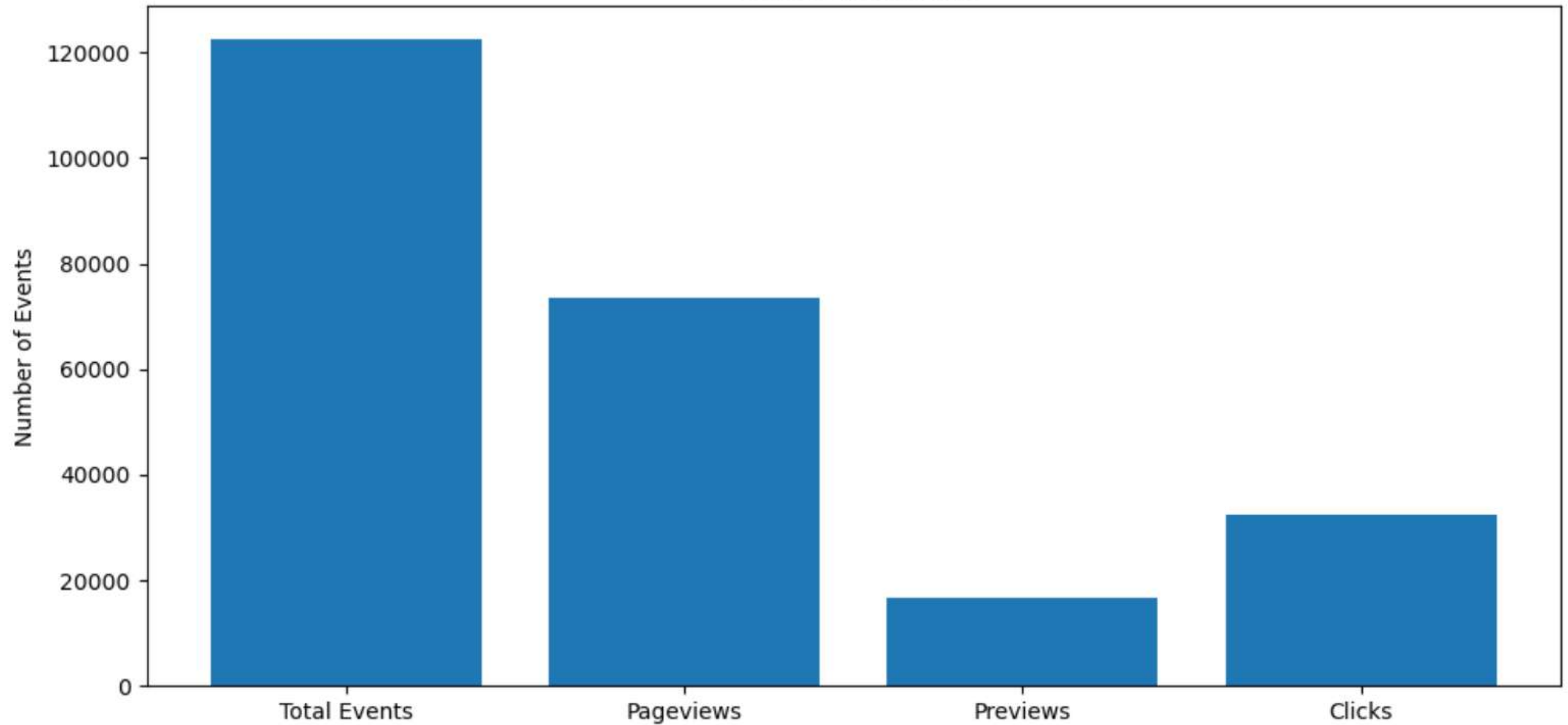
plt.title('Top Links by Event-Level Click Rate')
plt.xlabel('Click Rate (%)')
plt.ylabel('Link ID')

plt.tight_layout()
plt.show()
```



```
: kpi_values = [  
    total_events,  
    pageviews,  
    previews,  
    clicks  
]  
  
kpi_names = [  
    'Total Events',  
    'Pageviews',  
    'Previews',  
    'Clicks'  
]  
  
plt.figure(figsize=(10, 5))  
  
plt.bar(kpi_names, kpi_values)  
  
plt.title('Website Traffic KPIs')  
plt.ylabel('Number of Events')  
  
plt.tight_layout()  
plt.show()
```

Website Traffic KPIs



```
journey_summary = pd.DataFrame({  
    'Stage': [  
        'Pageview',  
        'Preview',  
        'Click'  
    ],  
    'Events': [  
        pageviews,  
        previews,  
        clicks  
    ]  
})
```

journey_summary

	Stage	Events
--	-------	--------

0	Pageview	73360
---	----------	-------

1	Preview	16708
---	---------	-------

2	Click	32499
---	-------	-------

```
traffic_summary = pd.DataFrame({
    'Metric': [
        'Total Events',
        'Pageviews',
        'Previews',
        'Clicks',
        'Pageview → Click Rate',
        'Pageview → Preview Rate',
        'Countries',
        'Cities',
        'Artists',
        'Tracks',
        'Links'
    ],
    'Value': [
        total_events,
        pageviews,
        previews,
        clicks,
        f"{pageview_click_rate:.2f}%",
        f"{pageview_preview_rate:.2f}%",
        df['country'].nunique(),
        df['city'].nunique(),
        df['artist'].nunique(),
        df['track'].nunique(),
        df['linkid'].nunique()
    ]
})
```

traffic_summary

	Metric	Value
0	Total Events	122567
1	Pageviews	73360
2	Previews	16708
3	Clicks	32499
4	Pageview → Click Rate	44.30%
5	Pageview → Preview Rate	22.78%
6	Countries	211
7	Cities	11993
8	Artists	2419
9	Tracks	3562
10	Links	3839

```
traffic_summary.to_csv(  
    'traffic_summary.csv',  
    index=False  
)  
  
daily_event.to_csv(  
    'daily_event_analysis.csv'  
)  
  
country_event_top.to_csv(  
    'country_event_analysis.csv'  
)  
  
artist_engagement.to_csv(  
    'artist_engagement.csv'  
)  
  
link_engagement.to_csv(  
    'link_engagement.csv'  
)  
  
print("Phase 3 analysis files saved successfully!")
```

Phase 3 analysis files saved successfully!